

Harish-Chandra Research Institute

Introduction to Algebraic Geometry

Assignment 3

M.Sc. Mathematics, Year II • August–December 2026

k will denote an algebraically closed field and $\mathbb{P}^n = \mathbb{P}^n(k)$ has homogeneous coordinates $[X_1 : \cdots : X_{n+1}]$. A projective change of coordinates is an automorphism induced by an element of $\mathrm{PGL}_{n+1}(k)$. For a projective variety V , write $k(V)$ for its function field and $\mathcal{O}_{V,P}$ for its local ring at P . If $C = V(F) \subseteq \mathbb{P}^2$ is a plane curve, $m_P(C) = m_P(F)$ denotes its multiplicity at P ; a point of multiplicity 1 is called simple.

Problem 1

A closed subset $L \subseteq \mathbb{P}^n$ is called *projectively linear* if $L = V(H_1, \dots, H_r)$ for homogeneous linear forms H_i . For $-1 \leq m \leq n$, put

$$\Lambda_{-1} = \emptyset, \quad \Lambda_m = V(X_{m+2}, \dots, X_{n+1}) \quad (0 \leq m < n), \quad \Lambda_n = \mathbb{P}^n.$$

- (a) If T is a projective change of coordinates, prove that $L^T := T^{-1}(L)$ is projectively linear.
- (b) Prove that $L^T = \Lambda_m$ for some projective change of coordinates T and some $m \in \{-1, 0, \dots, n\}$. Hence every nonempty projectively linear subset is isomorphic to $\mathbb{P}^m$ and is a projective variety.
- (c) Show that m is determined uniquely by L . Define $\dim L = m$, with the convention $\dim \emptyset = -1$.

Problem 2

Let $H_1, \dots, H_r$ be hyperplanes in $\mathbb{P}^n$, with $r \leq n$. Prove that

$$H_1 \cap \cdots \cap H_r \neq \emptyset.$$

Problem 3

For $z \in k(V)$ on a projective variety V , set

$$\mathrm{Reg}_V(z) = \{P \in V : z \in \mathcal{O}_{V,P}\}, \quad \mathrm{Pole}_V(z) = V \setminus \mathrm{Reg}_V(z).$$

Prove that $\mathrm{Reg}_V(z)$ is Zariski open, and hence that $\mathrm{Pole}_V(z)$ is an algebraic subset of V .

Problem 4

Let $(\mathbb{P}^n)^\vee$ be the dual projective space, with coordinates $[A_1 : \cdots : A_{n+1}]$. A hyperplane

$$H = V(a_1 X_1 + \cdots + a_{n+1} X_{n+1})$$

determines the nonzero vector $(a_1, \dots, a_{n+1})$ up to scalar.

- (a) Prove that $H \mapsto H^* := [a_1 : \cdots : a_{n+1}]$ is a well-defined bijection from the hyperplanes of $\mathbb{P}^n$ to the points of $(\mathbb{P}^n)^\vee$.
- (b) For $P = [p_1 : \cdots : p_{n+1}] \in \mathbb{P}^n$, define

$$P^* := V(p_1 A_1 + \cdots + p_{n+1} A_{n+1}) \subseteq (\mathbb{P}^n)^\vee.$$

Under the canonical double-dual identification, prove that $P^{**} = P$, $H^{**} = H$, and

$$P \in H \iff H^* \in P^*.$$

Problem 5

Let $V \subseteq W \subseteq \mathbb{P}^n$ be projective varieties, and suppose that V is a hypersurface in $\mathbb{P}^n$. Prove that $W = V$ or $W = \mathbb{P}^n$.

Problem 6

Assume that $\text{char}(k) \notin \{2, 3\}$. For each polynomial F below, prove that $V(F) \subseteq \mathbb{P}^2$ is irreducible. Determine every multiple point, its multiplicity, and its tangent lines, counted with multiplicity.

- (a) $F = XY^4 + YZ^4 + XZ^4$.
- (b) $F = X^2Y^3 + X^2Z^3 + Y^2Z^3$.
- (c) $F = Y^2Z - X(X - Z)(X - \lambda Z)$, where $\lambda \in k$.
- (d) $F = X^n + Y^n + Z^n$, where $n \geq 1$ and $\text{char}(k) \nmid n$.

Problem 7

Let $C = V(F) \subseteq \mathbb{P}^2$, where $F \in k[X, Y, Z]$ is homogeneous, and let $P \in C$ be simple. Prove that

$$T_PC = V(F_X(P)X + F_Y(P)Y + F_Z(P)Z).$$

Problem 8

Let $C, D \subseteq \mathbb{P}^2$ be plane curves with no common irreducible component. Prove that $C \cap D$ is finite.

Problem 9

Assume that $\text{char}(k) \neq 3$. Let $C = V(F) \subseteq \mathbb{P}^2$ be an irreducible cubic with an ordinary cusp at $P = [0 : 0 : 1]$, whose unique tangent line is $Y = 0$.

- (a) Prove that

$$F = aY^2Z - bX^3 - cX^2Y - dXY^2 - eY^3,$$

where $a, b \in k^\times$ and $c, d, e \in k$.

- (b) Prove that every irreducible cuspidal plane cubic is projectively equivalent to $Y^2Z = X^3$, and verify that this curve has no singular point other than $[0 : 0 : 1]$.